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ACTUATOR FOR THREE-DIMENSIONAL PRINTING SYSTEM

Abstract

A three-dimensional printing system includes a tank containing a liquid photopolymer resin. An arm is configured to be movable relative to the tank. A rigid base is connected to the arm. A light source is configured to emit light to the tank to form an object on the rigid base. An actuator is connected to at least one of the build plate and the tank. The actuator is configured to deform when power is supplied to the actuator.

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Background/Summary

BACKGROUND

Technical Field

[0001] The present disclosure generally relates to a three-dimensional printing system. More specifically, the present disclosure relates to an actuator for a three-dimensional printing system.

Background Information

[0002] Three-dimensional (3D) printing is the construction of a three-dimensional object from a digital file, such as a CAD model or a digital 3D model. The objects are printed layer by layer by the 3D printing system by curing portions of a light curable photopolymer resin layer by layer, one layer at a time, within a printing area of a tank filled with the photopolymer resin. A curing device, such as an ultraviolet light source, is projected through a transparent substrate, window or bottom wall of the tank curing each layer of the object on a build plate, or rigid base, that is initially at least partially submerged within the photopolymer resin. The build plate is incrementally raised upward as each layer is cured thereon.

[0003] When the build plate is raised to print the next layer, a resin void is left between the build plate and the transparent window. The resin void results in a lack of resin supply, which reduces a printing speed. When larger parts are being printed, the larger cross-sectional areas require an increased resin supply.

[0004] Printing a high strength polymer requires a high viscosity resin, which has a slower flow rate. The slower flow rate results in an increased amount of time required to fill the resin void, which reduces the printing speed.

SUMMARY

[0005] A need exists for an improved three-dimensional printing system in which an actuator is configured to deform when power is supplied thereto. The deformation of the actuator facilitates resin flow in the tank.

[0006] In view of the state of the known technology, one aspect of the present disclosure is to provide a three-dimensional printing system. The three-dimensional printing system includes a tank containing a liquid photopolymer resin. An arm is configured to be movable relative to the tank. A rigid base is connected to the arm. A light source is configured to emit light to the tank to form an object on the rigid base. An actuator is connected to at least one of the build plate and the tank. The actuator is configured to deform when power is supplied to the actuator.

[0007] Also other objects, features, aspects and advantages of the disclosed actuator for a three-dimensional printing system will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses exemplary embodiments of an actuator for a three-dimensional printing system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] Referring now to the attached drawings which form a part of this original disclosure:

[0009] FIG. 1 is an elevational view of a three-dimensional printing system including an actuator connected between an arm and a rigid base in accordance with an exemplary embodiment;

[0010] FIG. 2 is an elevational view of the three-dimensional printing system of FIG. 1 including a power source connected to the actuator;

[0011] FIGS. 3A and 3B are elevational views of an actuator before and after deformation, respectively;

[0012] FIG. 4 is an elevational view of a three-dimensional printing system of FIG. 1 in which each of the actuators is deformed;

[0013] FIG. 5 is an elevational view of a three-dimensional printing system of FIG. 1 in which one of a plurality of actuators is deformed;

[0014] FIG. 6 is a perspective view of an actuator disposed between a rigid base and an arm of the three-dimensional printing system of FIG. 1;

[0015] FIG. **7** is a top plan view of a rigid base including a first plurality of actuators;
[0016] FIG. **8** is a top plan view of a rigid base including a second plurality of actuators;
[0017] FIG. **9** is an elevational view of a three-dimensional printing system including an actuator connected to a tank in accordance with another exemplary embodiment;
[0018] FIG. **10** an elevational view of the three-dimensional printing system of FIG. **9** including a power source connected to the actuator;
[0019] FIG. **11** is an elevational view of the three-dimensional printing system of FIG. **9** in which each actuator is deformed;
[0020] FIG. **12** is an elevational view of a three-dimensional printing system of FIG. **9** in which one of a plurality of actuators is deformed; and
[0021] FIG. **13** is an elevational view of a three-dimensional printing system including a first actuator disposed between a rigid base and an arm and a second actuator connected to the tank in accordance with still another exemplary embodiment.

DETAILED DESCRIPTION OF EMBODIMENTS

[0022] Selected embodiments will now be explained with reference to the drawings. It will be apparent to those skilled in the art from this disclosure that the following descriptions of the embodiments are provided for illustration only and not for the purpose of limiting the invention as defined by the appended claims and their equivalents.

[0023] Referring initially to FIG. **1**, three-dimensional (3D) printing system **10** in accordance with an exemplary embodiment includes a tank **12** that contains a liquid photopolymer resin **14**. The 3D printing system **10** further includes a rigid base **16** on which an object **18** is to be printed and a control arm **20** connected to the rigid base **16**.

[0024] The rigid base **16** has a print surface **16A** on which the object **18** is configured to be printed, as shown in FIG. **1**. The control arm **20** is connected to the rigid base **16** to move the rigid base **16** relative to the tank **12**. A light source **22** is configured to emit light **24** to the tank **12** to form the printed object **18** on the rigid base **16**. An actuator **26** is connected to at least one of the rigid base **16** and the tank **12**. The actuator **26** is configured to deform when power is supplied to the actuator **26** from a power source **28**, as shown in FIG. **2**.

[0025] The tank **12** contains the liquid photopolymer resin **14**, as shown in FIG. **1**. The tank **12** can be any suitable shape to hold the liquid polymer resin **14** therein, such as rectangular or circular. The tank **14** has a base **30** and a side wall **32** extending upwardly from the base **30**. The base **30** is preferably transparent such that the light **24** emitted from the light source **22** passes through the base **30**. The entirety of the base **30** can be transparent, or a portion of the base **30** can be transparent. The transparent portion of the base **30** constitutes an optically transparent window **30A** through which the emitted light **22** is configured to pass.

[0026] The rigid base, or build platform, build plate or print bed, **16** provides the surface **16A** on which the object **18** is printed. The print surface **16A** is preferably a planar surface, as shown in FIG. **1**. The rigid base **16** can be made of any suitable material, such as plastic, such as polylactic acid (PLA), or glass. The rigid base **16** is connected to the control arm **20**.

[0027] The control arm **20** is connected to the rigid base **16** to control movement and positioning of the rigid base **16** during the printing process. The control arm **20** is connected to the rigid base **16** to move the rigid base **16** relative to the tank **12**. In other words, the arm **20** is configured to be movable relative to the tank **12** such that the movement of the arm **20** moves the rigid base **16**. The control arm **20** includes a plurality of links **20A**, **20B**, **20C** and **20D**, as shown in FIG. **6**, independently movable relative to each other to provide highly accurate positioning of the rigid base **16**. Although the control arm **20** is shown with four links **20A**, **20B**, **20C** and **20D**, the control arm **20** can have any suitable number of links **20**. The control arm **20** preferably has six degrees of freedom, such that the rigid base **16** can move through a curvilinear path to more accurately print the object **18**. The control arm **20** is preferably a robotic arm having six degrees of freedom. The six degrees of freedom are movements along the three axes (i.e., the X, Y and Z axes), and rotation

about each of the three axes (i.e., pitch, roll and yaw). Providing the control arm **20** with multiple degrees of freedom, such as six degrees of freedom, allows the control arm **20** to move the rigid base **16** through a curvilinear path, including moving the rigid base **16** to a plurality of positions, thereby allowing a more accurate object **18** to be printed.

[0028] The liquid polymer resin **14** is selectively cured by light-activated polymerization, such as by photopolymerization, which preferably uses visible or UV light, although light having any suitable wavelength can be used, to form in situ cross-linked polymer structures. The liquid polymer resin **14** preferably includes monomer and oligomer molecules that are converted to solid polymers during photopolymerization when the light **24** emitted by the light source **22** is guided through the transparent portion, or the optically transparent window **30A**, of the base **30** of the tank **12**.

[0029] The light source **22** emits light **24** to cure the liquid polymer resin **14** in the tank **12**, as shown in FIG. **1**. The light source **21** preferably emits UV light **24** having a wavelength between approximately 10 and 400 nanometers, inclusive. Preferably, the emitted UV light **24** has a wavelength between approximately 380 and 400 nanometers, inclusive. Light having any suitable wavelength can be used, such as, but not limited to, UV, visible and infrared light.

[0030] As shown in FIG. **1**, the printed object **18** is formed on the surface **16A** of the rigid base **16**. The printed object **18** is based on a model supplied to a computer (not shown) that controls the 3D printing process. The light **24** emitted from the light source **24** is guided to the tank **12** to cure the liquid polymer resin **14** on the surface **16A** of the rigid base **16** to form a first layer of the printed object **18**. The control arm **20** is connected to the rigid base **1** to move the rigid base **16** relative to the tank **12** in a direction indicated by arrow **34** away from the optically transparent window **30A** of the base **30**. The rigid base **16** is moved a distance approximately equal to a thickness of the formed layer. The light **24** is emitted from the light source **22** to cure the liquid polymer resin **14** in the tank **12** to form a second layer on the first layer. This process is repeated until the entire object is printed. When the printing is complete, the printed object **18** can be removed from the print surface **16A** of the rigid base **16**.

[0031] The transparent window **30A** of the base is optically transparent such that focused beams of light **24** emitted from the light source **22** pass therethrough and at predetermined areas or portions of the polymerizable resin **14** located within a printing area **36**. The printing area **36** is defined as the space below the rigid base **16** (at and/or below a lower surface **18A** of the object **18** being printed) and the upper surface **30B** of the transparent window **30A**.

[0032] As shown in FIGS. **1-8**, the actuator **26** is connected between the rigid base **16** and the arm **20**. The actuator **26** is preferably a piezoelectric stack, although the actuator can be any suitable actuator configured to move the rigid base **16** relative to the arm **20**. The piezoelectric stack **26** is a conventional structure including a plurality of piezo elements layered on top of one another to form the piezoelectric stack. The piezoelectric stack **26** is preferably configured to deform, or elongate, between approximately 30 and 100 microns, inclusive, when power is supplied to the piezoelectric stack. The piezoelectric stack **26** is preferably rated for a pull force between approximately 50 and 100 newtons, respectively.

[0033] The piezoelectric stack **26** is connected between an upper surface **16B** of the rigid base **16** and the control arm **20**, as shown in FIG. **1**, in any suitable manner, such as with an adhesive or by a threaded connection. The piezo elements of the piezoelectric stack **26** can be any suitable material including, but not limited to, lead zirconium titanate, barium titanate, and bismuth scandium lead titanate (BSPT, BiScO.sub.3—PbTiO.sub.3). The piezoelectric stack **26** is configured to provide precision deformation and a large force, such that the resulting movement of the rigid base **16** and/or the tank **12** can be controlled.

[0034] Any suitable number of piezoelectric stacks **26** can be connected between the rigid base **16** and the control arm **20**. As shown in FIG. **7**, four piezoelectric stacks **26** are connected between the upper surface **16B** of the rigid base **16** and the control arm **20** (FIG. **6**). As shown in FIG. **8**, six

piezoelectric stacks **26** are connected between the upper surface **16B** of the rigid base **16** and the control arm **20** (FIG. **6**). The piezoelectric stacks **26** are preferably disposed around an outer periphery of the upper surface **16B** of the rigid base **16** to provide greater movement of the rigid base **16** when deforming the piezoelectric stacks **26**. The number of the piezoelectric stacks **26** can be based on a final weight of the object **18** to be printed.

[0035] A power source **28** is electrically connected to each piezoelectric stack **26**, as shown in FIGS. **2-5**. Supplying a voltage to the piezoelectric stack **26** from the power source **28** causes a deformation of the piezoelectric stack **26**. The power source **28** is any suitable power source, such as a 120V battery. A controller **50** is connected to each power source **28** to control the supply of power to the piezoelectric stack **26**. Alternatively, a single controller **50** can be connected to all the piezoelectric stacks **26**.

[0036] As shown in FIG. **3A**, the piezoelectric stack **26** has a length L when a voltage is not supplied to the piezoelectric stack **26**. When a voltage is supplied from the power source **28** to the piezoelectric stack **26**, the piezoelectric stack **26** deforms by a length ΔL , such that the total length of the deformed piezoelectric stack **26** is $L + \Delta L$, as shown in FIG. **3B**. The deformation of the piezoelectric stack **26**, as shown in FIG. **3B**, is in a longitudinal direction of the piezoelectric stack. As shown in FIGS. **1, 2, 4** and **5**, the piezoelectric stack deforms in a first direction indicated by the arrow **40** when voltage is supplied, and retracts in a second direction indicated by an arrow **38** when the supply of voltage to the piezoelectric stack is stopped. In other words, supplying voltage deforms the piezoelectric stack by the longitudinal distance ΔL , and stopping the voltage supply returns the piezoelectric stack to the original, or default, length L . The piezoelectric stack **26** having the length L , as shown in FIG. **3A**, is a first position of the piezoelectric stack **26**. The piezoelectric stack **26** having the length $L + \Delta L$, as shown in FIG. **3B**, is a second position of the piezoelectric stack. Each of the plurality of piezoelectric stacks **26** is configured to deform a predetermined distance, i.e., ΔL , from a first position shown in FIG. **3A** to a second position shown in FIG. **3B** when power is supplied thereto. Each of the piezoelectric stacks **26** is configured to return from the second position (FIG. **3B**) to the first position (FIG. **3A**) when the supply of power is stopped. Each of the piezoelectric stacks **26** is configured to be independently controllable by having an independent power supply **28** connected thereto.

[0037] As shown in FIG. **1**, a suction force, indicated by arrow **44**, is generated between the printed object **18** and the base **20** of the tank **12**, which must be overcome to move the rigid base **16** upwardly to print the next layer on the printed object **18**. Printed objects **18** having a larger cross sectional area generate a larger suction force **44**. A resin void is generated in the print area **36** by the upward movement of the rigid base **16**. The resin **14** in the tank flows toward the resin void, as indicated by the arrows **42**. A lack of resin **14** quickly filling the resin void creates pressure on the printed object **18**, which can damage or negatively impact the printed object **18**. Decreased resin flow reduces the printing speed. The printed object **18** can also be damaged or otherwise negatively impacted by insufficient resin in the printing area **36** when printing the next layer of the object **18**. To print a high strength polymer, higher viscosity resins are used. The increased viscosity of the resin **14** further reduces the flow rate of the resin **14** to fill the resin void.

[0038] To offset the suction force and to increase the resin flow to the printing area **36**, the piezoelectric stacks **26** are activated to increase the resin flow **42** to the printing area **36**. The piezoelectric stacks **26** are initially in the second position. In other words, power is supplied from the power source **28** to the piezoelectric stacks **26** such that the piezoelectric stacks have a length of $L + \Delta L$ (i.e., the second position shown in FIG. **3B**). As shown in FIG. **2**, each of the piezoelectric stacks is maintained at the length $L + \Delta L$. ΔL can be any suitable length, such as, but not limited to, between approximately 30 and 100 microns, inclusive. The supply of power from the power source **28** to each of the piezoelectric stacks **26** is then stopped, such that each of the piezoelectric stacks **26** reduces in length by ΔL , as shown in FIG. **4**. The piezoelectric stacks **26** move in the direction indicated by arrow **38** in FIG. **4**. Power is then supplied to each of the piezoelectric stacks **26**, such

that that each of the piezoelectric stacks **26** increases in length by ΔL , as shown in FIG. 2. The piezoelectric stacks **26** move in the direction indicated by arrow **40** in FIG. 2. The retraction in length of the piezoelectric stacks **26** retracts the rigid base **16**, which induces resin flow to the printing area **36**. The subsequent extension of the piezoelectric stacks **26** pushes the rigid base **16** in the downward direction, further inducing the resin flow **42** to the printing area **36**. The oscillation, or extension and retraction of the piezoelectric stacks **26**, facilitates the resin flow **42** to the printing area **36** such that a resin void is not formed in the printing area **36** and sufficient resin is present to print the next layer of the object **18**. Each of the piezoelectric stacks **26**, such as the first and second piezoelectric stacks **26A** and **26B**, are controlled to simultaneously retract and elongate. In other words, the plurality of piezoelectric stacks **26** are deformed to facilitate resin flow toward the printing area **36** between the printed object **18** and the base **30** of the tank **12**.

[0039] When printing discrete layers, the piezoelectric stacks **26** can be retracted and extended once at the distance ΔL to facilitate resin flow **42** to the printing area **26**. When continuously printing the printed object **18**, the piezoelectric stacks **26** can be retracted and elongated at a predetermined frequency at a smaller distance (i.e., less than ΔL) to facilitate the resin flow **42** to the printing area **36**. Alternatively, when printing discrete layers of the printed object **18**, the piezoelectric stacks **26** can be retracted and elongated once at the distance ΔL to facilitate the resin flow **42** to the printing area **36**.

[0040] As shown in FIG. 5, each of the plurality of piezoelectric stacks **26** can be independently controlled. The first piezoelectric stack **26A** is retracted. The second piezoelectric stack **26B** is maintained in the elongated position. In other words, the supply of power from the power source **28** to the first piezoelectric stack **26A** is stopped such that the first piezoelectric stack **26A** has a length L , and the power source **28** supplies power to the second piezoelectric stack **26B** such that the second piezoelectric stack **26B** has a length $L + \Delta L$. As shown in FIG. 5, the first piezoelectric stack **26A** is in the first position, and the second piezoelectric stack **26B** is in the second position. Independently controlling each of the piezoelectric stacks **26**, as shown in FIG. 5, generates rotational movement of the rigid base, thereby increasing the resin flow **42** to the printing area **36**. Resin flow introduced by the rotation of the rigid base **16** flows relative to the stagnant resin in the tank **14**. The friction at the solid-liquid interface (between the base **30** of the tank **12** and the resin **14**) produces fluid movement to increase the resin flow **42** toward the printing area **36**.

[0041] The piezoelectric stacks **26** can be retracted and elongated at a predetermined frequency to induce shear stress in the resin **14**, as shown in FIG. 2. In other words, the piezoelectric stacks **26** are deformed at the predetermined frequency to generate the shear stress in the resin **14** to reduce a viscosity of the resin **14**. The shear stress can be used to reduce the viscosity of higher viscosity resins. Shear thinning is caused by the orderly movement of monomers in the resin caused by longer monomer chains aligning during movement of the resin.

[0042] The piezoelectric stacks **26** can also be deformed at a predetermined frequency to increase a temperature of the piezoelectric stacks **26**. The increased temperature of the piezoelectric stacks **26** increases the temperature of the resin **14** and the rigid base **16**. The increased temperature of the rigid base **16** further increases the temperature of the resin **14**, which reduces the viscosity of the resin. The piezoelectric stacks **26** can be deformed at the predetermined frequency prior to starting the printing process such that the resin **14** in the tank **12** is pre-heated to facilitate the resin flow during the printed process.

[0043] The deformation, or vibration, of the piezoelectric stacks **26** results in vibration within the printed object **18**. The vibration frequency of the piezoelectric stacks **26** can be tuned to the current mass of the printed object **18** for resonance vibration. During the printing process, the mass of the printed object **18** changes as each layer of the object **18** is printed. The vibration of the piezoelectric stacks **26** is changed to adjust for the increased mass of the printed object **18**. In other words, the predetermined frequency of the deformation of the piezoelectric stacks is based on a current mass of the printed object **18**. Adjusting the vibration frequency of the piezoelectric stacks

26 as the mass of the printed object **18** increases further facilitates resin flow, thereby increasing the printing speed.

[0044] A constant force is applied by the control arm **20** during the printing process. The piezoelectric stacks **26** generate electrical energy when displaced. The electrical energy generated by displacement of the piezoelectric stacks **26** can be monitored during the printing process. Variations in the constant force applied by the control arm **20** are indicated by the electrical energy generated by the piezoelectric stacks and reflect deviations from a mean force applied by the control arm **20**. The variations in the mean force indicate difficulty in removing the printed object from the base **30** of the tank **12** or in detachment of the printed object **18** from the rigid base **16**. A variation above a predetermined threshold of the mean force indicates that the printed object **18** is stuck to the base **30** of the tank **12**. A variation below the predetermined threshold of the mean force indicates that the printed object **18** has detached from the rigid base **16**. The printing process can be canceled when a variation outside the predetermined threshold is determined. The variations from the constant force applied by the control arm **20** determined by the piezoelectric stacks **26** can be used to determine replenishment intervals for the resin **14** and/or a lubricating oil disposed between the resin **14** and the base **30** of the tank **12**.

[0045] Deformation of the actuator **26**, as shown in FIGS. **1-8**, increases resin flow **24** in the tank **12** to the printing area **36** to increase printing speeds.

[0046] As shown in FIGS. **9-12**, a three-dimensional printing system **110** in accordance with another illustrated exemplary embodiment of the present invention is substantially similar to the 3D printing system **10** of the exemplary embodiment illustrated in FIGS. **1-8** except for the differences described below. Similar parts are identified with similar reference numerals, except increased by 100 (i.e., 1xx, accordingly).

[0047] An actuator **152** is connected to the tank **112**, as shown in FIGS. **9-12**. The actuator is preferably a piezoelectric stack **152**. Preferably, a plurality of piezoelectric stacks **152** are connected to the tank **112**.

[0048] A platform, or support, **146** is disposed beneath the tank **112**. The platform **146** can have an aperture through which the light **124** emitted by the light source **122** passes. Alternatively, the platform **146** can include a transparent portion through which the light **124** emitted by the light source **122** passes.

[0049] The plurality of piezoelectric stacks **152** are disposed between the base **130** of the tank **112** and the platform **146**. The plurality of piezoelectric stacks **152** are connected between an upper surface **146A** of the platform **146** and a lower surface **130C** of the base **130** of the tank **112**, as shown in FIGS. **9-12**. The piezoelectric stacks **152** are disposed between the light source **122** and the optically transparent window **130A** in a direction in which the light **124** is emitted. The piezoelectric stacks **152** are disposed between the light source **122** and the rigid base **116** in a direction **122** in which the light is emitted. The actuator **152** operates substantially similarly to the actuator **26** of FIGS. **1-8**.

[0050] A power source **154** is electrically connected to each piezoelectric stack **152**, as shown in FIGS. **10-12**. A controller **256** is electrically connected to each power source **254** to control the supply of power by the power source **254**. Supplying a voltage to the piezoelectric stack **152** from the power source **154** causes a deformation of the piezoelectric stack **152**. The power source **154** is any suitable power source, such as a 120V battery. A controller **156** is connected to each power source **154** to control the supply of power to the piezoelectric stack **152**. Alternatively, a single controller **156** can be connected to all the piezoelectric stacks **152**.

[0051] As shown in FIG. **10**, the piezoelectric stack **152** deforms in a first direction indicated by the arrow **158** when voltage is supplied from a power source **154**, and retracts in a second direction indicated by an arrow **160** when the supply of voltage to the piezoelectric stack **126** is stopped. In other words, supplying voltage deforms the piezoelectric stack **126** by the longitudinal distance ΔL , and stopping the voltage supply returns the piezoelectric stack to the original, or default, length L .

Deformation of the piezoelectric stacks **126** causes movement of the tank **112**, as shown in FIGS. **9-12**.

[0052] To offset the suction force **144** and to increase the flow of the resin **114** to the printing area **136**, the piezoelectric stacks **152** are activated to increase the resin flow **142** to the printing area **136**. The piezoelectric stacks **152** are initially in the second position, as shown in FIG. **10**. In other words, power is supplied from the power source **154** to the piezoelectric stacks **152** such that the piezoelectric stacks have a length of $L+\Delta L$ (i.e., the second position shown in FIG. **3B**). As shown in FIG. **10**, each of the piezoelectric stacks is maintained at the length $L+\Delta L$. ΔL can be any suitable length, such as, but not limited to, between approximately 30 and 100 microns, inclusive. Power from the power source **154** to each of the piezoelectric stacks **152** is then stopped, such that each of the piezoelectric stacks **152** reduces in length by ΔL , as shown in FIG. **11**. Each of the piezoelectric stacks **152** deforms in the direction indicated by arrow **160** in FIG. **11**, such that the tank **112** is moved closer to the platform **146**. Power is then supplied to each of the piezoelectric stacks **152**, such that each of the piezoelectric stacks **152** increases in length by ΔL , as shown in FIG. **10**. Each of the piezoelectric stacks **152** moves in the direction indicated by the arrow **158**. The retraction in length of the piezoelectric stacks **152** moves the tank **112**, which induces resin flow to the printing area **136**. The subsequent extension of the piezoelectric stacks **152** further moves the tank **112** in the upward direction, further inducing the resin flow **142** to the printing area **136**. The oscillation, or extension and retraction of the piezoelectric stacks **152**, facilitates the resin flow **142** to the printing area **136** such that a resin void is not formed in the printing area **136** and sufficient resin is present to print the next layer of the object **118**. Each of the piezoelectric stacks **152**, such as the first and second piezoelectric stacks **152A** and **156B**, are controlled to simultaneously retract and elongate. In other words, the plurality of piezoelectric stacks **152** are deformed to facilitate resin flow toward the printing area **136** between the printed object **118** and the base **130** of the tank **112**.

[0053] As shown in FIG. **12**, each of the plurality of piezoelectric stacks **152** can be independently controlled. The first piezoelectric stack **152A** is maintained in the elongated position. The second piezoelectric stack **152B** is retracted. In other words, the supply of power from the power source **154** to the second piezoelectric stack **152B** is stopped such that the second piezoelectric stack **152B** has a length L , and the power source **154** supplies power to the first piezoelectric stack **152A** such that the first piezoelectric stack **152A** has a length $L+\Delta L$. As shown in FIG. **12**, the first piezoelectric stack **152A** is in the second position, and the second piezoelectric stack **152B** is in the first position. Independently controlling each of the piezoelectric stacks **152**, as shown in FIG. **12**, generates rotational movement of the tank **112**, thereby increasing the resin flow **142** to the printing area **136**.

[0054] As shown in FIG. **13**, a three-dimensional printing system **210** in accordance with another illustrated exemplary embodiment of the present invention is substantially similar to the three-dimensional printing systems **10** and **110** of the exemplary embodiments illustrated in FIGS. **1-12** except for the differences described below. Similar parts are identified with similar reference numerals, except increased by 200 (i.e., 2xx, accordingly).

[0055] At least one first actuator **226** is connected to the rigid base **116** and at least one second actuator **252** is connected to the tank **212**, as shown in FIG. **13**. Preferably, a plurality of first piezoelectric stacks **226** are connected between the rigid base **216** and the control arm **220**, and a plurality of second piezoelectric stacks **252** are connected between a platform **246** and the base **230** of the tank **212**. The first and second piezoelectric stacks **226** and **252** operate substantially similar to the actuators **26** of FIGS. **1-8** and the actuators **152** of FIGS. **9-12**.

[0056] A power source **228** is electrically connected to each first piezoelectric stack **226**, as shown in FIG. **13**. A controller **250** is electrically connected to each power source **228** to control the supply of power by the power source **228**. A power source **254** is electrically connected to each piezoelectric stack **252**. A controller **256** is electrically connected to each power source **254** to

control the supply of power by the power source **254**.

[0057] Supplying a voltage to the first piezoelectric stack **226** from the power source **228** causes a deformation of the first piezoelectric stack **226**. The power source **228** is any suitable power source, such as a 120V battery. A controller **250** is connected to each power source **228** to control the supply of power to the first piezoelectric stack **226**. Alternatively, a single controller **250** can be connected to all the first piezoelectric stacks **226**.

[0058] Supplying a voltage to the second piezoelectric stack **252** from the power source **254** causes a deformation of the second piezoelectric stack **252**. The power source **254** is any suitable power source, such as a 120V battery. A controller **256** is connected to each power source **254** to control the supply of power to the second piezoelectric stack **252**. Alternatively, a single controller **256** can be connected to all the second piezoelectric stacks **252**.

[0059] As shown in FIG. **13**, the first piezoelectric stack **226** deforms in a first direction indicated by the arrow **240** when voltage is supplied from a power source **228**, and retracts in a second direction indicated by an arrow **238** when the supply of voltage to the first piezoelectric stack **226** is stopped. In other words, supplying voltage deforms the first piezoelectric stack **226** by the longitudinal distance ΔL , and stopping the voltage supply returns the first piezoelectric stack to the original, or default, length L . Deformation of the first piezoelectric stacks **226** causes movement of the rigid base **216**.

[0060] As shown in FIG. **13**, the second piezoelectric stack **252** deforms in a first direction indicated by the arrow **260** when voltage is supplied from a power source **254**, and retracts in a second direction indicated by an arrow **258** when the supply of voltage to the second piezoelectric stack **252** is stopped. In other words, supplying voltage deforms the second piezoelectric stack **252** by the longitudinal distance ΔL , and stopping the voltage supply returns the second piezoelectric stack to the original, or default, length L . Deformation of the second piezoelectric stacks **252** causes movement of the tank **212**.

[0061] To offset the suction force **244** and to increase the flow of the resin flow **214** to the printing area **236**, the first and second piezoelectric stacks **226** and **252** are activated to increase the resin flow **242** to the printing area **236**. As described above, the first and second piezoelectric stacks **226** and **252** are controllable to move the rigid base **216** and the tank **212**, respectively, to facilitate resin flow **242** to the printing area **236**.

General Interpretation of Terms

[0062] The controllers **50**, **154**, **250** and **254** preferably include a microcomputer with a control program that controls the supply of power from the power source as discussed above. The controllers can also include other conventional components, such as an input interface circuit, an output interface circuit, and storage devices, such as a ROM (Read Only Memory) device and a RAM (Random Access Memory) device. The microcomputer of the controllers are programmed to control the supply of power from the power sources. The memory circuit stores processing results and control programs, such as ones for the power supply operation and electrical field detection, that are run by the processor circuit. The controllers are operatively coupled to the power sources in a conventional manner. The internal RAM of the controllers store statuses of operational flags and various control data. The controllers are configured to selective control the power sources to control deformation of the actuators in accordance with the control programs.

[0063] In understanding the scope of the present invention, the term “comprising” and its derivatives, as used herein, are intended to be open ended terms that specify the presence of the stated features, elements, components, groups, integers, and/or steps, but do not exclude the presence of other unstated features, elements, components, groups, integers and/or steps. The foregoing also applies to words having similar meanings such as the terms, “including”, “having” and their derivatives. Also, the terms “part,” “section,” “portion,” “member” or “element” when used in the singular can have the dual meaning of a single part or a plurality of parts. Also as used herein to describe the above embodiment(s), the following directional terms “forward”, “rearward”,

“above”, “downward”, “vertical”, “horizontal”, “below” and “transverse” as well as any other similar directional terms refer to those directions of a three-dimensional printing system equipped with the actuator. Accordingly, these terms, as utilized to describe the present invention should be interpreted relative to a three-dimensional printing system equipped with the actuator.

[0064] The term “detect” as used herein to describe an operation or function carried out by a component, a section, a device or the like includes a component, a section, a device or the like that does not require physical detection, but rather includes determining, measuring, modeling, predicting or computing or the like to carry out the operation or function.

[0065] The terms of degree such as “substantially”, “about” and “approximately” as used herein mean a reasonable amount of deviation of the modified term such that the end result is not significantly changed.

[0066] While only selected embodiments have been chosen to illustrate the present invention, it will be apparent to those skilled in the art from this disclosure that various changes and modifications can be made herein without departing from the scope of the invention as defined in the appended claims. For example, the size, shape, location or orientation of the various components can be changed as needed and/or desired. Components that are shown directly connected or contacting each other can have intermediate structures disposed between them. The functions of one element can be performed by two, and vice versa. The structures and functions of one embodiment can be adopted in another embodiment. It is not necessary for all advantages to be present in a particular embodiment at the same time. Every feature which is unique from the prior art, alone or in combination with other features, also should be considered a separate description of further inventions by the applicant, including the structural and/or functional concepts embodied by such feature(s). Thus, the foregoing descriptions of the embodiments according to the present invention are provided for illustration only, and not for the purpose of limiting the invention as defined by the appended claims and their equivalents.

Claims

1. A three-dimensional printing system comprising: a tank containing a liquid photopolymer resin; an arm configured to be movable relative to the tank; a rigid base connected to the arm; a light source configured to emit light to the tank to form an object on the rigid base; and an actuator connected to at least one of the build plate and the tank, the actuator being configured to deform when power is supplied to the actuator.
2. The three-dimensional printing system according to claim 1, wherein the actuator is connected between the build plate and the arm.
3. The three-dimensional printing system according to claim 2, wherein the actuator is a plurality of piezoelectric stacks.
4. The three-dimensional printing system according to claim 3, wherein each of the plurality of piezoelectric stacks is configured to deform a predetermined distance from a first position to a second position when power is supplied thereto.
5. The three-dimensional printing system according to claim 4, wherein each of the plurality of piezoelectric stacks is configured to return from the second position to the first position when the supply of power is stopped.
6. The three-dimensional printing system according to claim 3, wherein each of the plurality of piezoelectric stacks is independently controllable.
7. The three-dimensional printing system according to claim 3, wherein the plurality of piezoelectric stacks are deformed to facilitate resin flow toward a printing area between the object and a base of the tank.
8. The three-dimensional printing system according to claim 3, wherein the plurality of piezoelectric stacks are deformed at a predetermined frequency to generate a shear stress in the

resin to reduce a viscosity of the resin.

9. The three-dimensional printing system according to claim 8, wherein the predetermined frequency is based on a current mass of the object.

10. The three-dimensional printing system according to claim 3, wherein the plurality of piezoelectric stacks are deformed at a predetermined frequency to increase a temperature of the resin.

11. The three-dimensional printing system according to claim 3, wherein electrical energy generated by the plurality of piezoelectric stacks is monitored to determine adhesion of the object to the tank.

12. The three-dimensional printing system according to claim 3, wherein a number of the plurality of piezoelectric stacks is based on a final weight of the object to be printed.

13. The three-dimensional printing system according to claim 1, wherein the actuator is connected to the tank.

14. The three-dimensional printing system according to claim 13, wherein the actuator is a plurality of piezoelectric stacks.

15. The three-dimensional printing system according to claim 14, wherein the plurality of piezoelectric stacks are disposed between a platform and a base of the tank.

16. The three-dimensional printing system according to claim 15, wherein each of the plurality of piezoelectric stacks is configured to move a predetermined distance from a first position to a second position when power is supplied thereto.

17. The three-dimensional printing system according to claim 15, wherein each of the plurality of piezoelectric stacks is configured to return from the second position to the first position when the supply of power is stopped.

18. The three-dimensional printing system according to claim 15, wherein each of the plurality of piezoelectric stacks is independently controllable.

19. The three-dimensional printing system according to claim 14, wherein the plurality of piezoelectric stacks are deformed to facilitate resin flow toward a printing area between the object and a base of the tank.

20. The three-dimensional printing system according to claim 14, wherein the plurality of piezoelectric stacks are disposed between the light source and the rigid base.
